Chapter 3

Gate-Level Minimization

1. Simplify the following Boolean functions, using the K- maps and implement them with two-level AND-OR gates:

   $F(x, y, z) = \sum (2, 3, 4, 5)$

   $F (x, y, z) = \sum (1, 2, 3, 5, 6, 7)$

   $F = A'B'C'D' + AC'D' + B'CD' + A'BCD + BC'D$

   $F = WXY + XZ + WX'Z + W'X$

2. Simplify the following Boolean functions F, together with the don't-care conditions d

   a) $F (x, y, z) = \sum m (0, 1, 4, 5, 6)$; $d (x, y, z) = \sum (2, 3, 7)$

   b) $F (A, B, C, D) = \sum m (0, 6, 8, 13, 14)$; $d (A, B, C, D) = \sum (2, 4, 10)$

   c) $F(A,B,C,D) = \sum m(1,3,5,7,9)$; $d(6,12,13)$

3. Simplify the following function F using K-map

   $F (w, x, y, z) = \sum(1, 2, 3, 5, 7, 15)$ and

   don't care $(w, x, y, z) = \sum(4, 6, 12, 13)$